

The inverse of the root element $X_{-1}(u)$ in the group $G(\mathfrak{m})$

Abstract

We give a self-contained account of a machine-checked proof of Corollary 3.2 of the source: in the group $G(\mathfrak{m})$, given by generators and relations, one has $X_{-1}(u)^{-1} = X_{-1}(-u)$ for every $u \in \mathbb{C}$. The formalization does not fix the coefficients $c(j)$ of the modular function J . Instead it treats $c: \mathbb{N} \rightarrow \mathbb{N}$ as an arbitrary parameter, so the result is proved for every such c , with no hypothesis on c . The proof uses three lemmas: defining relators become trivial in the presented group, the additivity relation (Re:1a) holds there, and $X_{-1}(0) = 1$. These lemmas and the final theorem are all machine-verified.

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Status	proof machine-verified (Lean 4, Mathlib)
Lemmas	3
Text	written by claude-opus-5-5, checked against the Lean source
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1 Setting

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$, and $\mathbb{C}^\times$ denotes the group of nonzero complex numbers. Fix an arbitrary function $c: \mathbb{N} \rightarrow \mathbb{N}$. In the source, $c(j)$ is the coefficient of q^j in $J(q)$. In the present text c is a free parameter, and nothing is assumed about it.

Definition 1.1 (Index set and structure constants). Let

$$E = E_c = \{(\ell, j, k) \in \mathbb{N}^3 \mid \ell < j, 1 \leq k \leq c(j)\}.$$

For $(\ell, j, k) \in E$ put

$$c_{\ell j} = (-1)^{\ell+1} \binom{j-1}{\ell} (\ell+1)(j-\ell) \in \mathbb{C}.$$

Here $j-1 \geq 0$ and $j-\ell \geq 1$, because $\ell < j$.

Definition 1.2 (Generators). Let $\mathcal{X}$ be the set of pairwise distinct symbols

$$H_1(s), H_2(s) \ (s \in \mathbb{C}^\times), \quad X_{-1}(u), Y_{-1}(u) \ (u \in \mathbb{C}), \quad X_{\ell,jk}(u), Y_{\ell,jk}(u) \ ((\ell, j, k) \in E, \ u \in \mathbb{C}).$$

Let $F(\mathcal{X})$ be the free group on $\mathcal{X}$.

For $a, b \in F(\mathcal{X})$ we write $(a, b) = aba^{-1}b^{-1}$. For $s \in \mathbb{C}^\times$ and $(\ell, j, k) \in E$ we set

$$\begin{aligned} \tilde{w}_{-1}(s) &= X_{-1}(s) Y_{-1}(-s^{-1}) X_{-1}(s), & \tilde{w}_{-1} &= \tilde{w}_{-1}(1), \\ \tilde{w}_{\ell,jk}(s) &= X_{\ell,jk}(s) Y_{\ell,jk}\left(\frac{-s^{-1}}{c_{\ell j}}\right) X_{\ell,jk}(s), & \tilde{w}_{\ell,jk} &= \tilde{w}_{\ell,jk}(1). \end{aligned}$$

Definition 1.3 (Relators and the group $G(\mathfrak{m})$). The set $\mathcal{R} \subseteq F(\mathcal{X})$ is defined from the relations listed below.

- A relation of the form $a = b$ contributes the relator ab^{-1} .
- A relation of the form $(a, b) = 1$ contributes the relator (a, b) .

The relations range over all $s, t \in \mathbb{C}^\times$, all $u, v \in \mathbb{C}$, and all $(\ell, j, k), (m, p, q) \in E$, subject to the side conditions stated.

$$H_1(s)H_1(t) = H_1(st), \tag{H:1a}$$

$$H_2(s)H_2(t) = H_2(st), \tag{H:1b}$$

$$H_1(s)H_2(t) = H_2(t)H_1(s), \tag{H:2}$$

$$X_{-1}(u)X_{-1}(v) = X_{-1}(u+v), \tag{Re:1a}$$

$$Y_{-1}(u)Y_{-1}(v) = Y_{-1}(u+v), \tag{Re:1b}$$

$$Y_{-1}(-t)X_{-1}(s)Y_{-1}(t) = X_{-1}(-t^{-1})Y_{-1}(-t^2s)X_{-1}(t^{-1}), \tag{Re:2}$$

$$\tilde{w}_{-1}(s)\tilde{w}_{-1} = H_1(-s)H_2(-s^{-1}), \tag{Re:3}$$

$$\tilde{w}_{-1}X_{-1}(u)\tilde{w}_{-1}^{-1} = Y_{-1}(-u), \tag{Re:4a}$$

$$\tilde{w}_{-1}Y_{-1}(u)\tilde{w}_{-1}^{-1} = X_{-1}(-u), \tag{Re:4b}$$

$$\tilde{w}_{-1}H_1(s)\tilde{w}_{-1}^{-1} = H_2(s), \tag{Re:5a}$$

$$\tilde{w}_{-1}H_2(s)\tilde{w}_{-1}^{-1} = H_1(s), \tag{Re:5b}$$

$$H_1(s)X_{-1}(u)H_1(s)^{-1} = X_{-1}(su), \tag{Re:6a}$$

$$H_2(s)X_{-1}(u)H_2(s)^{-1} = X_{-1}(s^{-1}u), \tag{Re:6b}$$

$$H_1(s)Y_{-1}(u)H_1(s)^{-1} = Y_{-1}(s^{-1}u), \tag{Re:6c}$$

$$H_2(s)Y_{-1}(u)H_2(s)^{-1} = Y_{-1}(su), \tag{Re:6d}$$

$$X_{\ell,jk}(u)X_{\ell,jk}(v) = X_{\ell,jk}(u+v), \tag{Im:1a}$$

$$Y_{\ell,jk}(u)Y_{\ell,jk}(v) = Y_{\ell,jk}(u+v), \tag{Im:1b}$$

$$Y_{\ell,jk}(-t)X_{\ell,jk}(s)Y_{\ell,jk}(t) = X_{\ell,jk}\left(\frac{-t^{-1}}{c_{\ell j}}\right)Y_{\ell,jk}(-c_{\ell j}t^2s)X_{\ell,jk}\left(\frac{t^{-1}}{c_{\ell j}}\right), \tag{Im:2}$$

$$\tilde{w}_{\ell,jk}(s^{(\ell+1)(j-\ell)})\tilde{w}_{\ell,jk} = H_1((-s)^{j-\ell})H_2((-s)^{\ell+1}), \tag{Im:3}$$

$$\tilde{w}_{\ell,jk}X_{\ell,jk}(u)\tilde{w}_{\ell,jk}^{-1} = Y_{\ell,jk}\left(\frac{-u}{c_{\ell j}}\right), \tag{Im:4a}$$

$$\tilde{w}_{\ell,jk}Y_{\ell,jk}(u)\tilde{w}_{\ell,jk}^{-1} = X_{\ell,jk}(-c_{\ell j}u), \tag{Im:4b}$$

$$\tilde{w}_{\ell,jk} H_1(s^{j-\ell}) \tilde{w}_{\ell,jk}^{-1} = H_2(s^{-(\ell+1)}), \quad (\text{Im:5a})$$

$$\tilde{w}_{\ell,jk} H_2(s^{\ell+1}) \tilde{w}_{\ell,jk}^{-1} = H_1(s^{-(j-\ell)}), \quad (\text{Im:5b})$$

$$H_1(s) X_{\ell,jk}(u) H_1(s)^{-1} = X_{\ell,jk}(s^{\ell+1}u), \quad (\text{Im:6a})$$

$$H_2(s) X_{\ell,jk}(u) H_2(s)^{-1} = X_{\ell,jk}(s^{j-\ell}u), \quad (\text{Im:6b})$$

$$H_1(s) Y_{\ell,jk}(u) H_1(s)^{-1} = Y_{\ell,jk}(s^{-(\ell+1)}u), \quad (\text{Im:6c})$$

$$H_2(s) Y_{\ell,jk}(u) H_2(s)^{-1} = Y_{\ell,jk}(s^{-(j-\ell)}u), \quad (\text{Im:6d})$$

$$(X_{-1}(u), X_{j-1,jk}(v)) = 1 \quad \text{whenever } (j-1, j, k) \in E, \quad (\text{U:1a})$$

$$(Y_{-1}(u), Y_{j-1,jk}(v)) = 1 \quad \text{whenever } (j-1, j, k) \in E, \quad (\text{U:1b})$$

$$(Y_{-1}(u), X_{0,jk}(v)) = 1 \quad \text{whenever } (0, j, k) \in E, \quad (\text{U:1c})$$

$$(X_{-1}(u), Y_{0,jk}(v)) = 1 \quad \text{whenever } (0, j, k) \in E, \quad (\text{U:1d})$$

$$(X_{\ell,jk}(u), Y_{m,pq}(v)) = 1 \quad \text{if } j \neq p \text{ or } k \neq q \text{ or } |\ell - m| > 1, \quad (\text{U:2})$$

$$(X_{1,2k}(u), Y_{0,2k}(v)) = X_{-1}(uv) \quad \text{whenever } (1, 2, k), (0, 2, k) \in E, \quad (\text{U:3a})$$

$$(X_{0,2k}(u), Y_{1,2k}(v)) = Y_{-1}(uv) \quad \text{whenever } (0, 2, k), (1, 2, k) \in E, \quad (\text{U:3b})$$

$$\tilde{w}_{-1} X_{\ell,jk}(u) \tilde{w}_{-1}^{-1} = X_{j-1-\ell,jk}((-1)^{j-\ell-1}u), \quad (\text{U:4a})$$

$$\tilde{w}_{-1} Y_{\ell,jk}(u) \tilde{w}_{-1}^{-1} = Y_{j-1-\ell,jk}((-1)^{j-\ell-1}u). \quad (\text{U:4b})$$

In (U:4a) and (U:4b), the triple $(j-1-\ell, j, k)$ lies in E automatically. Let $N_{\mathcal{R}}$ be the normal closure of $\mathcal{R}$ in $F(\mathcal{X})$. Define

$$G(\mathbf{m}) = G_c = \langle \mathcal{X} \mid \mathcal{R} \rangle = F(\mathcal{X})/N_{\mathcal{R}},$$

and let $\pi: F(\mathcal{X}) \rightarrow G_c$ be the quotient map. For a generator $x \in \mathcal{X}$ we also write x for its image $\pi(x)$ when no confusion can arise.

Theorem 1.4. *Let $c: \mathbb{N} \rightarrow \mathbb{N}$ be any function, and let $u \in \mathbb{C}$. Then in G_c*

$$\pi(X_{-1}(u))^{-1} = \pi(X_{-1}(-u)).$$

This is Corollary 3.2 of the source. The source takes c to be the coefficient function of J ; that is one instance of the parameter c here. The formal statement holds for every c , with no hypothesis such as $c(2) \geq 1$. It therefore contains the source statement as a special case.

2 Lemmas

Lemma 2.1 (Relators are trivial). *Let $c: \mathbb{N} \rightarrow \mathbb{N}$ be any function, and let $r \in F(\mathcal{X})$ be a word with $r \in \mathcal{R}$. Then $\pi(r) = 1$ in G_c .*

Proof. We have $r \in \mathcal{R} \subseteq N_{\mathcal{R}} = \ker \pi$. □

Lemma 2.2 (Additivity of X_{-1}). *Let $c: \mathbb{N} \rightarrow \mathbb{N}$ be any function, and let $u, v \in \mathbb{C}$. Then in G_c*

$$X_{-1}(u) X_{-1}(v) = X_{-1}(u+v).$$

Proof. By (Re:1a), the word $X_{-1}(u) X_{-1}(v) X_{-1}(u+v)^{-1}$ belongs to $\mathcal{R}$. By [Lemma 2.1](#),

$$\pi(X_{-1}(u)) \pi(X_{-1}(v)) \pi(X_{-1}(u+v))^{-1} = 1.$$

Multiplying on the right by $\pi(X_{-1}(u+v))$ gives the claim. □

Lemma 2.3 ($X_{-1}(0)$ is trivial). *Let $c: \mathbb{N} \rightarrow \mathbb{N}$ be any function. Then $X_{-1}(0) = 1$ in G_c .*

Proof. Take $u = v = 0$ in (Re:1a). The relator is then $X_{-1}(0)X_{-1}(0)X_{-1}(0+0)^{-1}$. Since $0+0=0$, this word equals

$$X_{-1}(0) X_{-1}(0) X_{-1}(0)^{-1} = X_{-1}(0)$$

in $F(\mathcal{X})$. Hence $X_{-1}(0) \in \mathcal{R}$, and [Lemma 2.1](#) gives $\pi(X_{-1}(0)) = 1$. $\square$

3 Proof of the theorem

Proof of [Theorem 1.4](#). In any group, if $ab = 1$ then $a^{-1} = b$. It therefore suffices to show $X_{-1}(u)X_{-1}(-u) = 1$ in G_c . Apply [Lemma 2.2](#) with $v = -u$, then use $u + (-u) = 0$ and [Lemma 2.3](#):

$$X_{-1}(u)X_{-1}(-u) = X_{-1}(u + (-u)) = X_{-1}(0) = 1.$$

Hence $X_{-1}(u)^{-1} = X_{-1}(-u)$. $\square$

The argument follows the source proof step for step. The source first rewrites (Re:1a) in relator form, which corresponds to [Lemma 2.1](#), and then in equational form, which is [Lemma 2.2](#). It then sets $v = -u$. The one difference concerns the fact $X_{-1}(0) = 1$. The source obtains it by citing its Lemma 3.1. Here it is [Lemma 2.3](#), proved directly from (Re:1a) at $u = v = 0$. Apart from this, the formal result is more general in one respect only: it holds for an arbitrary parameter $c: \mathbb{N} \rightarrow \mathbb{N}$ and does not need the specific coefficients of J .

A Correspondence with the formal proof

Each lemma of the text and the Lean declaration that states it; [Theorem 1.4](#) is `gsimple_x_inverse`.

Lemma	Lean declaration	Status
Lemma 2.1	<code>mk_rel_eq_one_in_G</code>	verified
Lemma 2.2	<code>Xm_add_in_G</code>	verified
Lemma 2.3	<code>Xm_zero_in_G</code>	verified